

ABSTRACT

Students and teachers do faces issues to access data from outside department; most likely faculty notes are needed and can only be accessible within their department. If teachers want to update some data from outside the department i.e. from home, so it is not possible for them to access the faculty notes. So they use other cloud applications every teacher chooses their own service, which is difficult for students to bookmark or remember every teacher's links of shared folders.

With the advancement of micro systems technologies reason behind this to unite the students and teachers on the single platform on which both teachers and students can access their files and folders from anywhere so that the teachers can upload their files and folders and their class related lectures, books and other stuff and the students can download their course related content uploaded by their respected teachers.

ACKNOWLEDGEMENT

In the name of Allah, the most Gracious and the Most Merciful.

Peace and blessing of Allah be upon Prophet Muhammad ﷺ

First, praise of Allah, for giving us this opportunity, the strength and the patience to complete our FYP finally, after the challenges and difficulties. We would like to thank our supervisor _____ for his guidance, motivation and most his significant contribution in this project, expert _____ and _____ for giving us the opportunity to work on this project. We would also like to thanks our parents for financial and moral support and our friends who have helped and motivated us throughout. May Allah reward them all abundantly. Ameen

DEDICATION

This report is dedicated to PAF-KIET University, our Teacher, our Supervisor, our Parents, our fellow colleagues and the hard-working students of PAF-KIET, with a hope that they will succeed in every aspect of their Academic Career and this project may help them in any aspect of their life.

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CHAPTER 1

1. Introduction

1.1. Motivations

Artificial intelligence, or AI, is replacing humans in tasks, and in the years to come, this will have a significant impact on the world. Health is not overlooked even as AI advances banking, research, education, and other fields. The traditional healthcare system desperately needs to change, and tech and API support can help bring about a revolution in this regard. We have combined .NET Core, HTML, CSS, JavaScript, React, and other technological elements into our FYP (Final Year Project) to create a more reliable, streamlined, and authentic web application. With AI and our expert system, Lifeline is easy for users to use and provides access to medical facilities to patients of all ages and doctors of several hospitals.

Despite the World Health Organization's (WHO) insight that e-health is a cost-effective, secure, reliable means of accessing medical care, little has changed in Pakistan. In this sense, Lifeline can serve as a one-stop shop for patients' needs as well as hospitals'.

1.2. Problem Statement

Patients' personal and health information is essential to their care, and hospitals keep track of it in their records. But, soft copies are always necessary to be within reach of patients at all times. Millions of lives can be saved thanks to archived medical records on smartphones and other devices. In addition, human error is a common occurrence in Pakistan's healthcare system as a result of misprescribed medications, delayed reports, late or missed treatments, staffing shortages, and other factors.

Yet in the field, AI support and the appropriate system enabled by API technology can be lifesavers. It can simultaneously provide support to millions of people and advise them about the most beneficial options—something that a helpline cannot do for all the patients at the same time.

1.3. Objectives and Contributions

The following are the objectives of Lifeline:

- With the support of chatbots, maps, and dashboard resources, users of all ages can operate it more easily.
- Millions of people who require medical attention can be reached. The AI-integrated API system can provide patients with real-time, knowledgeable assistance about what treatment they need and where to get it.
- Complete range of patient medical information, including Prescription History, Medical Portfolio, Appointment Notifications, and Emergency Contact Integration, are all integrated into the dashboard.
- Provides a clear system at every level, such as the employee, doctor, and hospital dashboards, to prevent errors and poor management.

The following are going to be the contributions of Lifeline in the health sector:

- Introducing chatbots can help users become more accustomed to using the system, which will make it easier for them to get medical help when they need it.
- It's also a safer way for patients and doctors to communicate, regardless of how far apart they are.
- Help at all levels so that doctors, patients, and the hospital administration are all aware of the latest status.

1.4. Project Scope

Lifeline encourages its users to have autonomy and provides them with information about everything related to their health with reports, notifications, and a streamlined system. Patients would be guided by the chatbot to the appropriate doctor and facility even if they are completely unaware of their condition, saving both time and lives. Furthermore, a complete web application that incorporates .NET Core, HTML, CSS, JavaScript, React, RL (Reinforcement Learning), and GNN (General Neural Network) has the potential to significantly change the data collection and management system in the health industry.

1.5. Organization of the Report

A structured overview of our Final Year Project "Lifeline" is provided in this report. The reasons, problem statement, objectives, and contributions will all be covered in detail in the parts that follow. Subsequent sections will cover the technical aspects of the project,

addressing its features and capabilities. A summary of the project's scope and a plan for further work are also included in the report.

CHAPTER 2

2. Literature Review/Process Review

2.1. Introduction

The existence of health apps and websites is undoubtedly evidence of the substantial amount of work that has been done in the field of health. The simplest interface, which provides users with instant access to health information, is essential for any user who is a patient. What could be more suitable in this instance than AI chatbots? Nevertheless, there is a lack of them in the user's web-based application systems. The problem is providing patients with a straightforward solution that is either impracticable or difficult to use, which causes users to give up on it quickly.

Doctors and hospital administration are overrun with work, and little is done to lighten their workload. However, things can be made easier for both of them with their individual dashboards, where they can view all the information about their patients, including their appointments and updated status of tasks and patients.

2.2. Literature Review

The healthcare industry is becoming increasingly automated these days, as evidenced by the numerous websites and applications that are available to help patients on the Internet. But, most of them are for data management or e-health facilities, hence, the apps are patient- or doctor-centric and even when integrate both, AI is nowhere around. Olufemi Ayodeji Odeniyi, A. Alabi, Anthonia, A. Kayode, and O. Makinde (2016) show considerable work in their telemedicine and e-healthcare mobile app. But, with its virtual communication system for kids and adults, it only concentrates on bridging the gap between patients and physicians. Only half of the issue is resolved by the mobile app system, which is limited to consultation and is powered by Java 2 'Micro Edition' (J2ME) and Net Beans IDE 6.9.

Al-Ghrai, Mohammed, and Saeed (2021) created a website that allow convenience for patients as well as doctors to access medical records and information in one location. Doctor appointments, billing, test and treatment schedules, medication information, and other

patient-related data are all managed by the database. Without a doubt, the website helps to lessen the effort and chances for errors when using the SQL Server, ASP MVC5, and C# programming language. However, there is no integration of AI and API on the website.

2.3. Functional and Non Functional Requirements

2.3.1. *Functional Requirements*

Lifeline is a web application that is perfect for users of all ages because all of its features complement one another. The API of the chatbot on Flask and .Net Core assistance stores records for a seamless and timely medical experience, including the Medical Portfolio, Blood Availability Monitoring, Appointment Notifications, Prescription History, and more. Easy and detailed information sharing between patients, doctors, and medical staff is made possible by the SQL Server database.

Using Google Maps, the chatbot first directs the patient to the closest and best location for treatment, as well as the hospital's or clinic's address. On the other hand, the dashboard provides the hospital and doctor with clear information about the system. The time of the appointment as well as other patient information can be added by hospital staff members to the dashboards of their individual doctors. Including this, the physician has access to all information on their dashboard, which involves everything related to patient records, hospital services, and their own schedule.

With Lifeline, users can see the hospital's facilities as well as a specific directory of doctors, staff members, and other details. In addition, patients who are confined to bed, live far away, or are unable to see the doctor in person can schedule a video-call appointment, during which the doctor can virtually check them and even prescribe medications and tests.

2.3.2. *Non-Functional Requirements*

For a web-based application, performance is crucial. The system's ease of use contributes to increasing its efficiency. One feature that improves scalability and provides assistance to multiple users simultaneously without causing errors is the real-time informed help. Once the patient is aware of the treatment, has access to reports, and can notified for everything then they don't have to worry about anything other than their health.

2.4. Project Significance

Lack of access to healthcare in Pakistan puts millions of lives at risk for a number of reasons, including a lack of understanding a lack of technological integration in the healthcare system, expensive treatment, and delayed care. With the use of the web-based application's Chatbot and Search Bar features, Lifeline can quickly and effectively provide users with genuine assistance for many of the aforementioned problems. In addition, individuals can receive medical care at home and avoid travel expenses by using the virtual treatment center.

A clear picture of the patients' health issues can be obtained, regardless of their location, through the medical records and portfolios that doctors and hospital employees enter for each patient in the dashboard. The technologically advanced Lifeline system can save lives and save time for patients, doctors, and management alike.

2.5. Software Platform

- Visual Studio
- Visual Studio Code
- Microsoft SQL Server Management Studio

2.6. Scalability

The Lifeline system has been developed using the HTML, CSS, and JS programming languages. Additionally, the SQL server enables the system to store and analyze the data of multiple users. The use of .Net Core runs the system smoothly no matter how many people use the system at once, Flask for chatbot API and Net Core for web API allow it to function seamlessly. Not only that, but Google Maps also helps with hospital location finding.

2.7. Services

The following services are offered by Lifeline to its users.

- Real-Time Informed Help
- Prescription History
- Medical Portfolio
- Appointment Notifications
- Blood Availability Monitoring
- Emergency Contact Integration

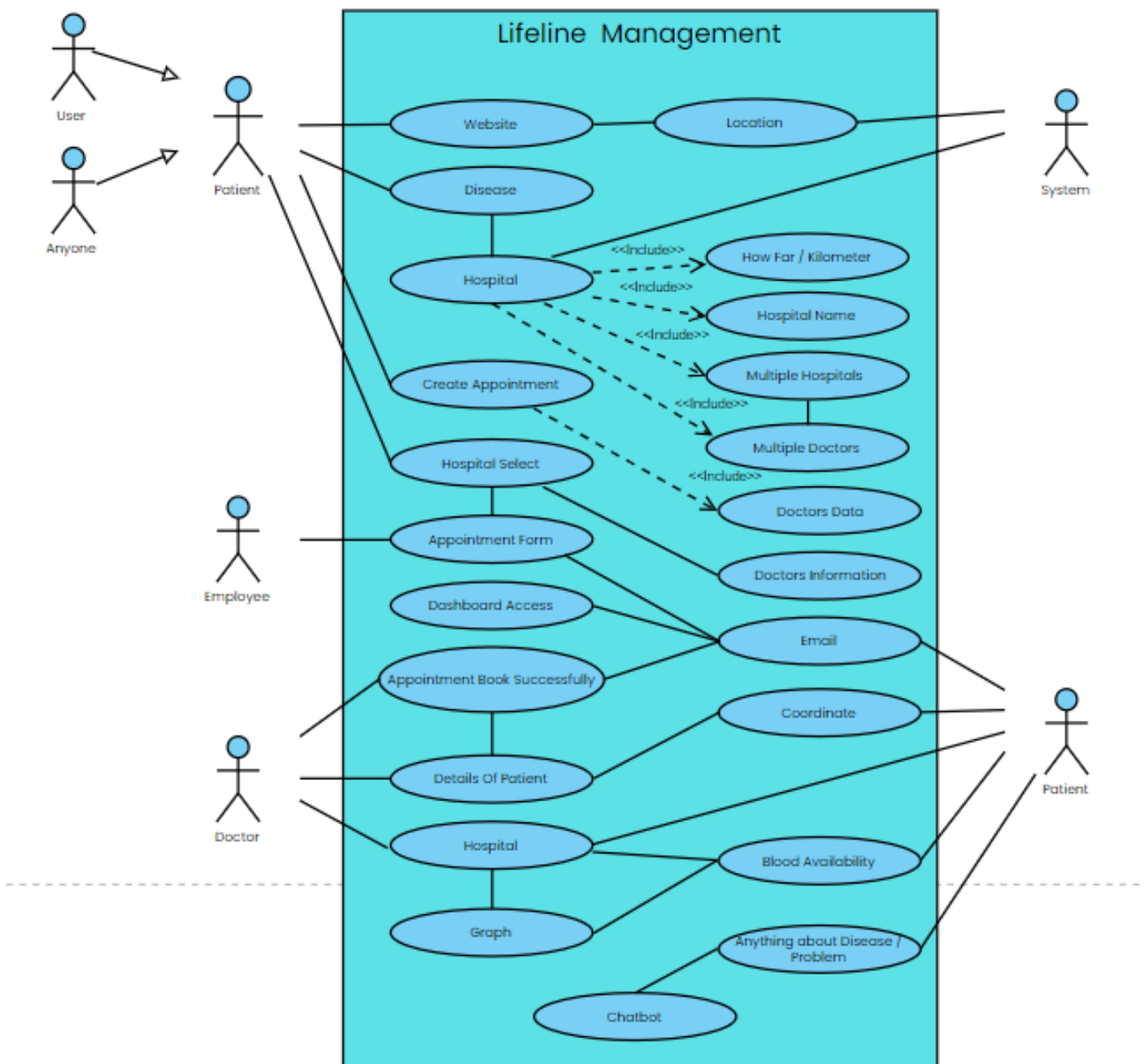
- Admin Dashboard
- Hospital Dashboard
- Patient Dashboard
- Doctor Dashboard
- Employee Dashboard

CHAPTER 3

3. Projects diagrams

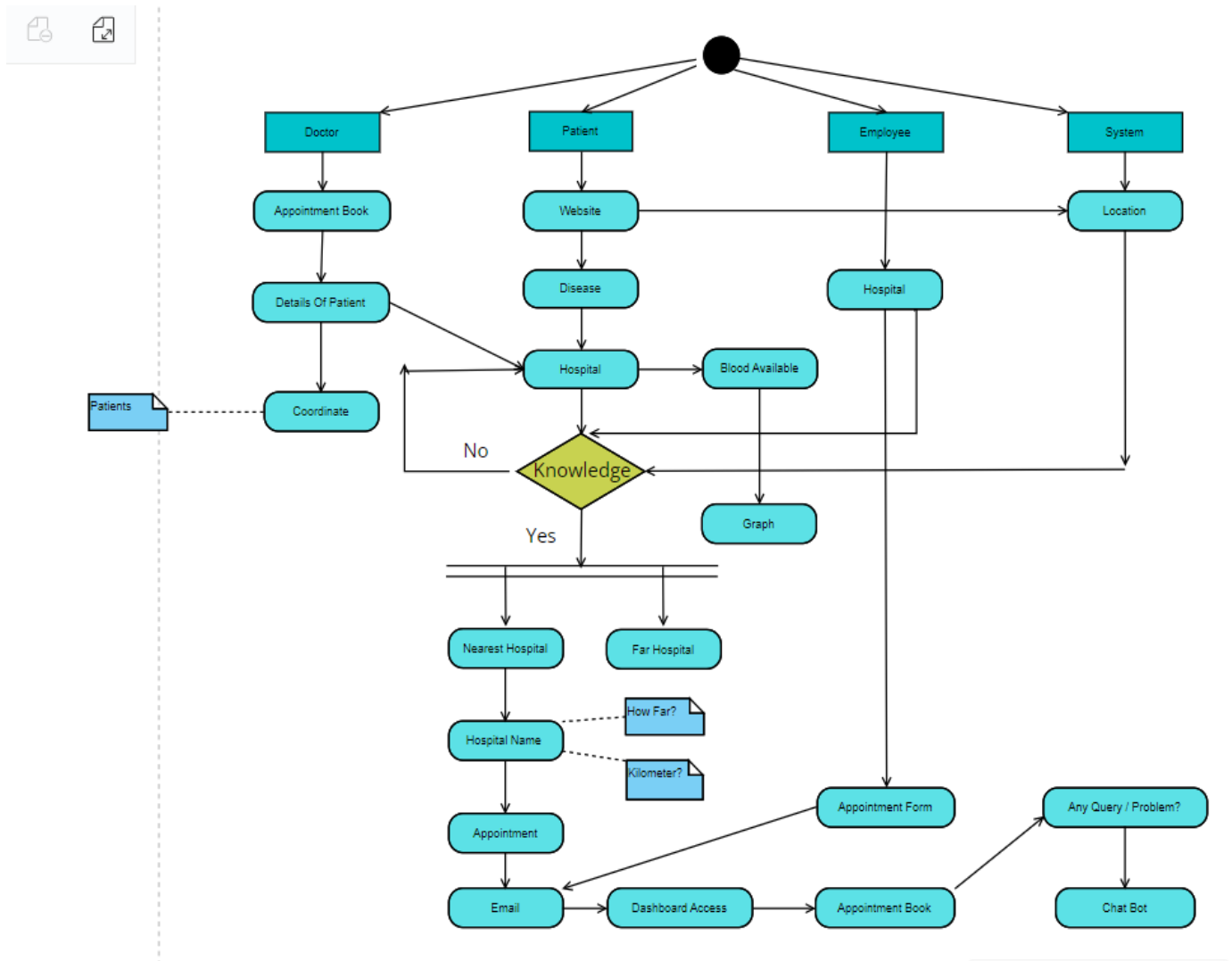
Based on the above literature review and project scope here are some diagrams, which illustrates that what will be our project or the system is capable to reach the desired results.

3.1. Use Case Diagrams

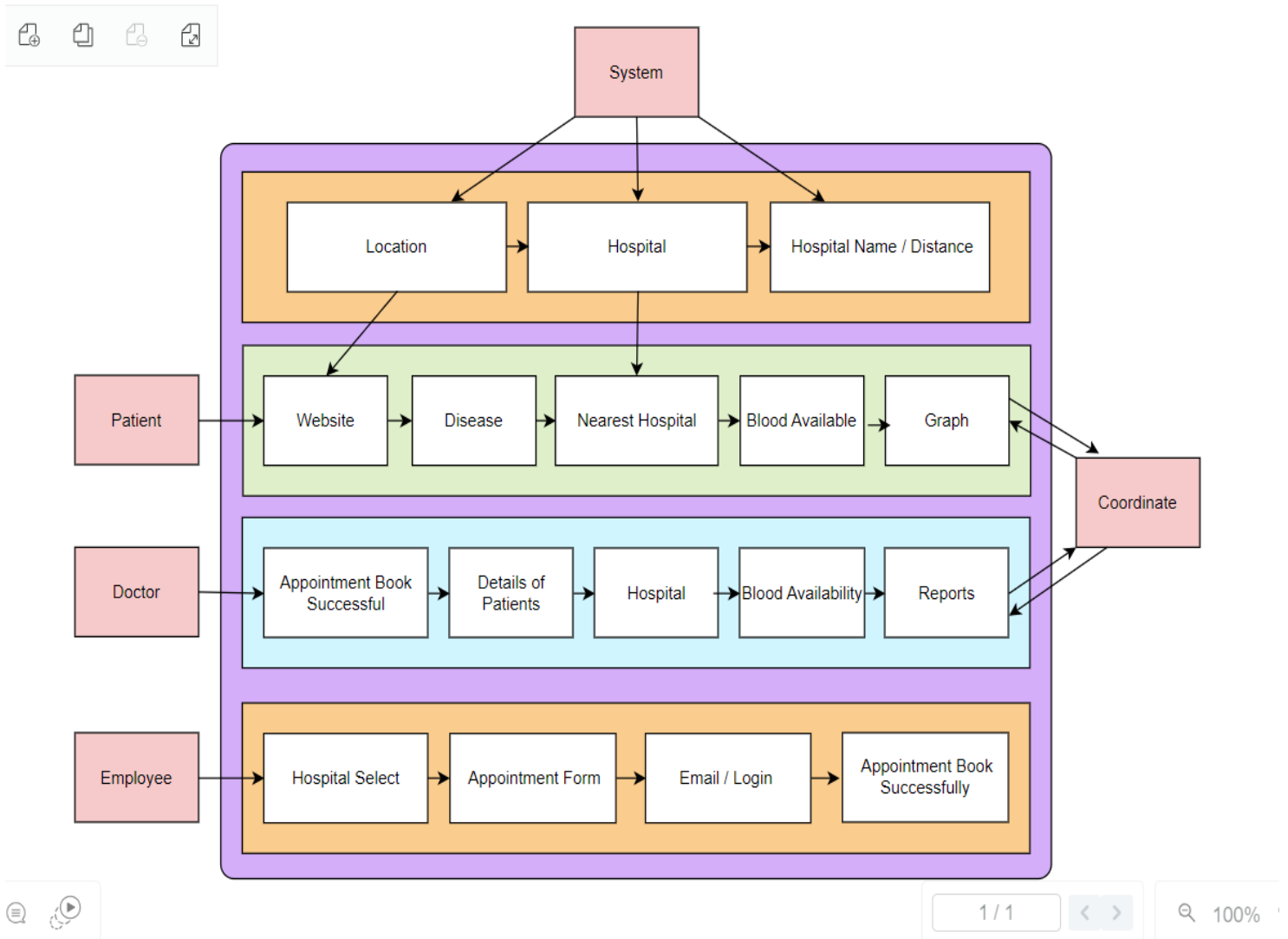


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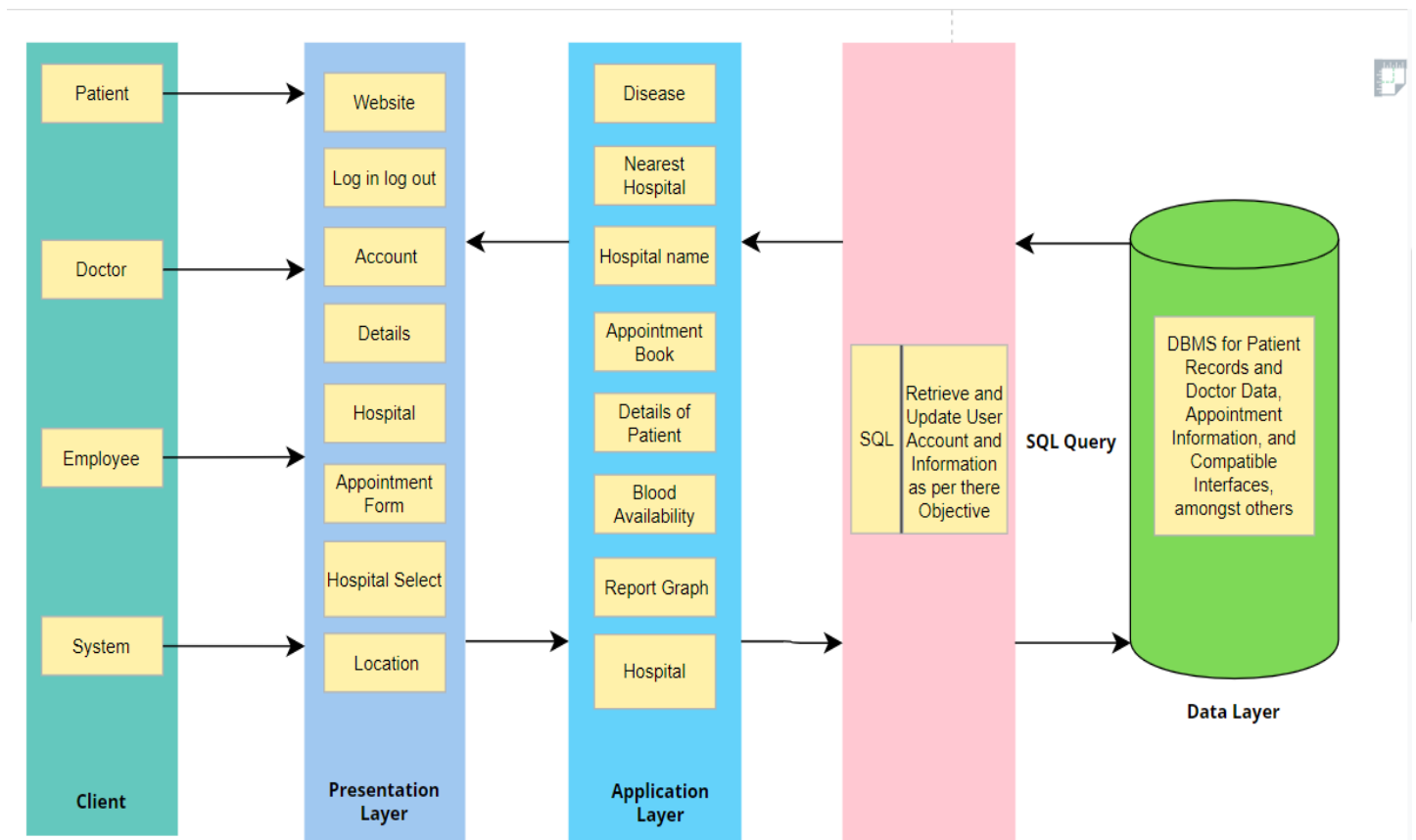
3.2. Activity diagram (diagram correction)



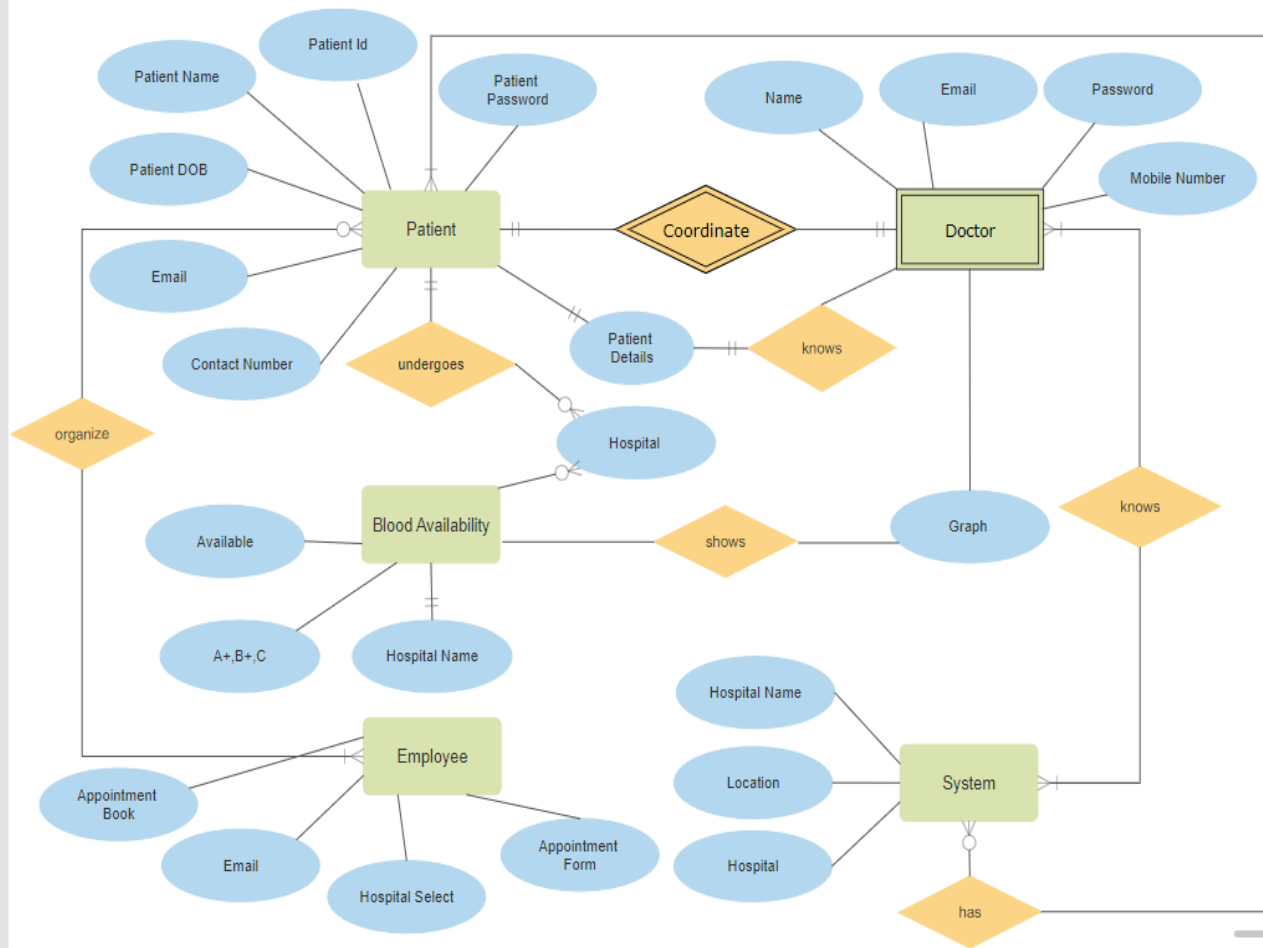
3.3. System Block Diagram



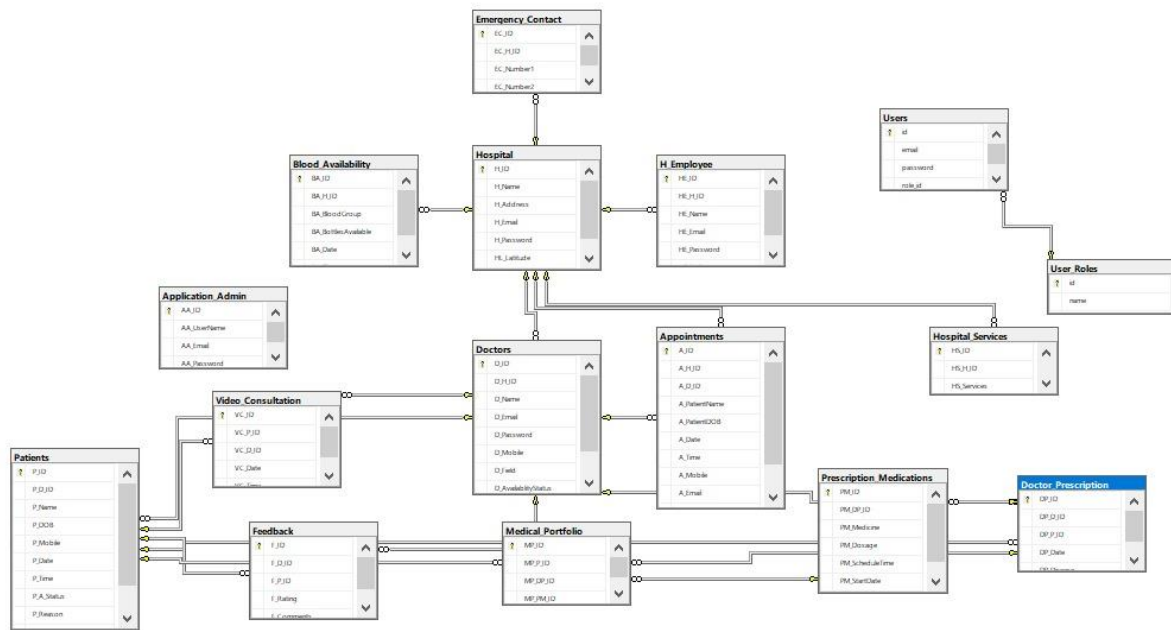
3.4. System Architecture



3.5. ER Diagram



3.5.1. ERD Table Dictionary



3.6. Inside Project

3.6.1. Python Flask API

The Python Flask API functions as the backend framework, employing Flask to establish restful endpoints, enabling the handling of HTTP requests and responses. It ensures seamless communication with databases managing incoming requests to provide essential data and services to clients. We used this to connect our trained Chabot with our website.

3.6.2. .NET Core Web API

The .NET Core Web API serves as a backend system built on .NET Core, furnishing endpoints for client server interactions. Using the .NET Core framework, it establishes controllers to define routes and handle incoming HTTP requests. This API delivers functionalities and data access to frontend applications or external systems seamlessly integrating with databases and external services. Our project's complete backend relies on this technology.

3.6.3. React

React for Frontend, a JavaScript library, specializes in creating user interfaces especially for single page applications. Leveraging the React library it constructs reusable UI components

and efficiently manages their states. With the virtual DOM optimizing rendering it empowers the system by crafting responsive interactive interfaces managing user interactions states and rendering for the frontend application. Our project's complete UI is built using this library.

3.6.4. Html and CSS

HTML and CSS form the bedrock of web development. HTML structures the content and elements of web pages while CSS defines their presentation and styling. Their significance lies in providing the foundation and aesthetics of the application. HTML organizes content, delineating layout elements while CSS enhances visual appeal and styling. Together they create user friendly and visually pleasing interfaces complementing the functionalities provided by backend and frontend frameworks.

3.7. Used Technologies

3.7.1. Visual Studio

Visual Studio is a powerful integrated development environment IDE widely used by developers. It serves as a comprehensive tool kit offering tools and features essential for software development. Its purpose is to provide a platform for debugging, testing etc.

3.7.2. Visual Studio Code

Visual Studio Code is a lightweight yet powerful source code editor used by developers. Its renowned for its versatility, supporting multiple programming languages and offering an extensive library of extensions.

3.7.3. Microsoft SQL Server Management Studio

Microsoft SQL Server Management Studio is a robust and comprehensive tool used for managing SQL Server databases. Its primary purpose is to provide a user friendly interface for database administrators and developers to efficiently interact with and manage SQL Server databases.

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